## INTRODUCTION

This instruction outlines the field engineering procedures to be follow when installing the Gas Turbine 125 Volt Battery System in "FA" and advanced "H" series gas turbines.

These instructions also provide general instructions for battery system installation for the MS 3001, MS 5001, MS 5002, MS 6001, (and MS 7001, MS 9001 three-bearing set) Gas Turbine units. (Consult the manufacturer's battery system specific instructions for installation details on these units).

The installation procedures provided in this article cover the arrival on site, unpacking, inspection, rigging, storage rack assembly, installation and startup of C&D (wet 1.250 specific gravity calcium alloyed lead acid battery systems) installed into storage racks.

The battery system location is in the (PEECC) Packaged Electrical Electronic Control Center (MLI A059) for most "FA" series gas turbines. Batteries supplied for "H" class units, or single shaft 7001FA and 9001FA STAG units are installed in the plant battery room. (See Battery Room Equipment).

## REFERENCE DOCUMENTS

| No. | Description |
|---|---|
| C&D Manual 12-800 | C&D Power Systems Service & Installation Manual |
| C&D Manual RS-759 | ARE Series Single–Phase Input |
| MLI-0330 | Battery Installation Instructions |
| MLI-1207 | Battery Arrangements Outline |
| DN 523 | MS 9000H Advanced Gas Turbine Installation Guide |

## GENERAL INFORMATION

The applicable vendor Service Instructions will normally be found in the GE Service Manual under the Station Auxiliaries & Services Tab. The Battery Arrangement Outline drawing (ML 1207) provides battery dimensions, layout and connections. A copy of C&D Manual 12-800 (Standby Battery Flooded Cell Installation & Operating Instruction). Is also included in the “Accessories Package” shipped with each battery system.

### Purpose of Battery Systems

Station auxiliaries and services consist of power distribution components used to provide auxiliary power for starting and operating the gas turbine, compartment lighting, heating and cooling requirements, DC emergency power and standby power needs.

During normal gas turbine operation, control and power needs are furnished by the turbine driven load generator using the equipment described. At the time of initial startup and during shutdown periods, standby AC power from some other source (customer supplied) is required to maintain the gas turbine in a state of readiness.

Standby AC power is required to insure the immediate startup capability of particular turbine equipment and related control systems when the start signal is given. Standby power is required for the following functions:

1. Heating and circulating turbine lube oil at low ambient temperatures to maintain proper oil viscosity.
2. Control panel heating.
3. Generator space heating.
4. Compartment heating.
5. Battery charging.
6. Exhaust frame blower motors.
7. Operation of control compartment air conditioners during periods of high ambient temperatures to keep electrical equipment insulation within design limits.

**SAFETY REQUIREMENTS**

**WARNING: Shock hazards are present — DO NOT TOUCH uninsulated battery, connectors or terminals.**

**WARNING: Gas produced by these batteries can be explosive. Protect eyes when working around battery. Provide adequate ventilation so that hydrogen gas accumulations in the battery areas do not exceed two (2) percent. DO NOT SMOKE, USE OPEN FLAME or CREATE SPARKS near battery.**

**WARNING: These batteries contain sulfuric acid which can cause severe burns. In case of skin contact with Electrolyte, immediately remove contaminated clothing and flush affected areas thoroughly with water. If eye contact has occurred, flush for a minimum of 15 minutes with large amounts of running water and seek immediate medical attention.**

All appropriate persons should be made aware of these safety warnings. Eye wash facilities must be immediately available. As a minimum, at least one quart of potable water in a plastic squeeze bottle is recommended. Facilities for additional flushing of eyes and skin should be nearby. If eye contact has occurred, flush for a minimum of 15 minutes with large amounts of running water and seek immediate medical attention.

Before working on a battery, workers should remove any static charge from his/her person or any object, by touching a grounded metal surface.

Use safety equipment such as the following (when appropriate to the site and/or required by Federal, State or local laws):

1. Safety glasses (or face shield).
2. Macron polyester lab coat (or acid resistant shirt and pants).
3. Finger fitting PVC gloves and insulated tools.
4. A portable (or fixed) eye wash station and/or shower may be required.

## SPECIAL TOOLS

Because of the possibility of explosion, it is strongly recommended that non-sparking bronze wrenches be used on this equipment.

## RECEIVING AND STORAGE

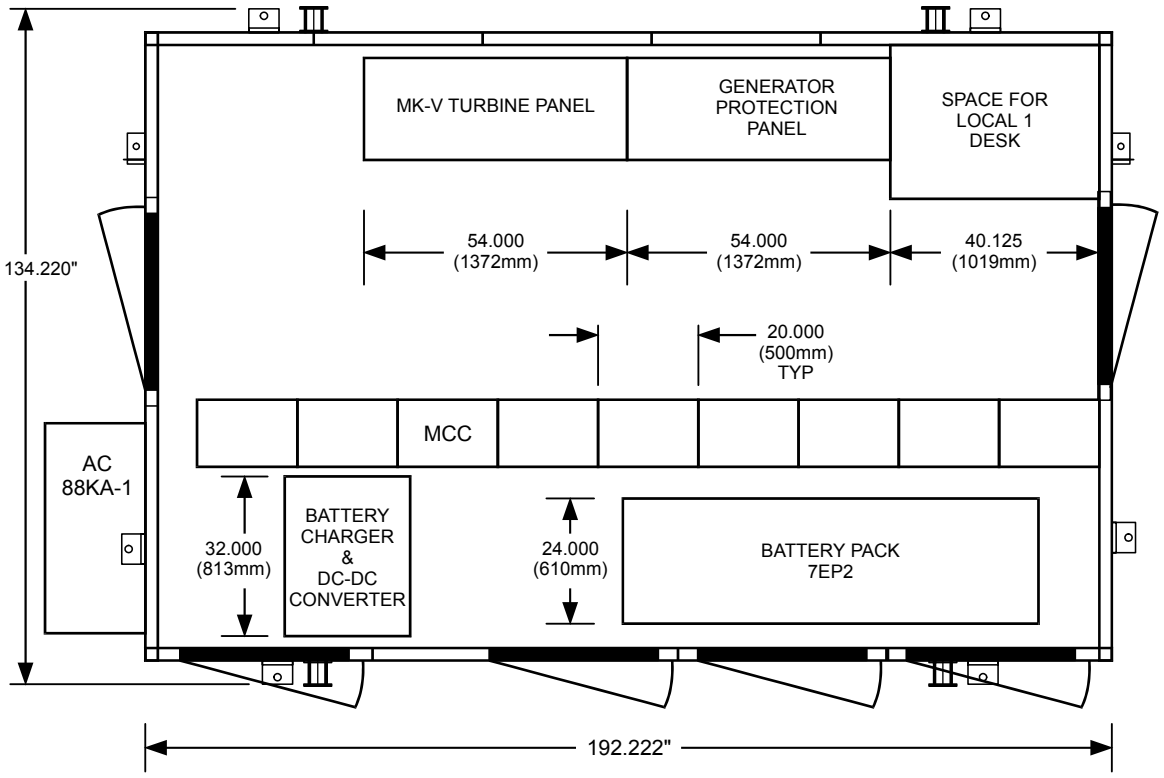
Refer to C&D Installation Manual 12-800 for receiving and storage instructions. If a copy of this instruction is not available before arrival of batteries, retrieve the copy shipped in the Accessories Package as soon as practical. (See Fig. 1).

Read the Receiving and Storage sections and the Initial Charge section: General (especially the last paragraph) of C&D 12-800 Manual before accepting and/or handling the battery. (See Figs. 2 and 3).

### General Information

Every precaution has been taken to pack the battery for shipment to assure its safe arrival. As soon as the battery is received, check the packing material for evidence of damage in transit. If the packing material is physically damaged or wet acid stains are present, make a notation on the Bill of Lading before the shipment is signed for.

**Note:** For F.O.B. factory shipments, shipping damage must be handled directly with the carrier, not with C&D Charter Power Systems.




**PLAN VIEW**

GT61B070

**Figure 1. MS 61B Packaged Electrical Electronic Control Center (MLI A059) – Typical**




**Figure 2. Battery Packing for Shipment – Typical**




**Note:** Batteries on shipping pallet, moved next to PEECC with forklift (foreground)

**Figure 3. Battery Racks Inside PEECC**

### Unpacking and Handling

Large single cell and multicell monobloc batteries are packed in individual cartons and strapped to wooden pallets for shipping. (See Fig. 2). Remove the straps and carefully lift off the cartons for access to lifting slots in the bottom support pieces. (See Fig. 4).

Always lift cells from the bottom. **Never lift cells by the posts.** A lifting sling and spreader are packed in the accompanying accessories box. Slip the lifting sling under the cell, then install the spreader block on top of the cell. Use the loops in the sling to hoist the cell. (See Fig. 5).

**Note:** Lifting sling loops must be crossed or held together above the spreader block, (see Fig. 5), before lifting a battery unit (4-cell jar) to prevent the jar from tipping and/or falling from the sling before placement on the rack.

### Pre-Installation Procedures

At the first opportunity, check the electrolyte level in each cell. It should be between the “high” and “low” level lines on the container. If the level is more than 1/2 inch below the “high” mark, **make no water additions until the cells have been on a "float" charge for one week.** (Contact the local C&D agent when this condition exists). Fill cells with a low level to the same level as the other cells in the battery using electrolyte taken from cells with a high level.

### Accessories Package

Check the contents of the Accessories Package against the enclosed packing list to confirm that all connecting hardware, terminal plates (if ordered), and cables have been received. Place accessories in a safe place from which they can be easily retrieved when needed. Record shortages and contact the local C&D agent, as required.

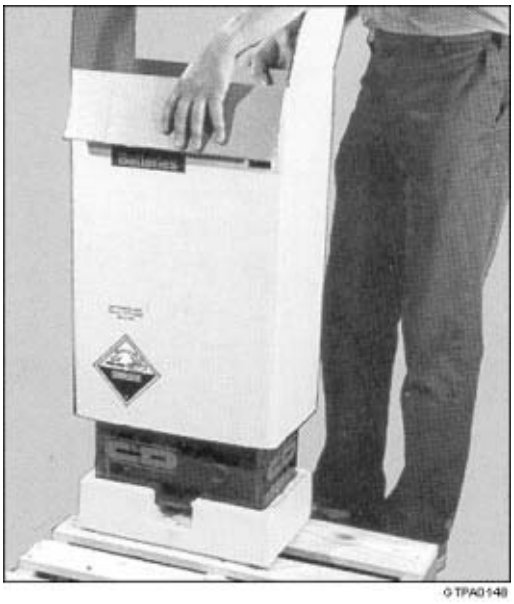



**Figure 4. Removing Battery Packing Materials**




**Figure 5. Use of Lifting Sling and Spreader Block**

### Battery Rack

There are two types of battery racks, (Non-seismic and Earthquake protected). The following describes assembly of both racks:

### Standard Rack Assembly (Non-Seismic)

1. Check all parts received against the Bill of Materials on the applicable C&D drawing. **(**Do not assemble unless all materials are at job site).
2. Mark frame locations on the floor, install anchor bolts (if required) and position frames over bolts.
3. Install braces onto frames (Fig. 6), using appropriate bolts, lock washers and nuts. Hand-tighten only.

4. Install rails when all frames are in proper upright position. Hand-tighten only.

   **Note:** When installing racks more than 10 feet long use appropriate bolts, lock washers and nuts. (See Spliced Rail Installation Detail in Fig. 6).

5. Check frames to verify that they are in the correct vertical position, and the assembly is level in all directions. Then torque bolts/nuts to nuts to the appropriate value (typically 25-30 ft. lbs. for 3/8-16 bolts). Tighten the braces first, then rails. When the assembly is complete, bolt the frames to the floor.

6. Place insulating rail covers (rubber mats) onto rails, if they are not already installed.

**Earthquake-Protected (EP) Racks**

1. Check all parts received against the bill of material on the C&D drawing. (Do not assemble unless all material is at the job site).

2. Mark frame locations on the floor, install anchor bolts (if required), and position frames over bolts.

3. Install braces onto frames (Fig. 7), using appropriate bolts, lock washers and nuts. Hand-tighten only.

4. Install bottom rails and rear side rails (not front side rails) when all frames are in the proper upright position using appropriate bolts, lock washers and nuts. Hand-tighten only.

5. Check frames to verify they are in the correct vertical position and that the assembly is level in all directions. Then, torque bolts/nuts to the appropriate value (typically 25-30 ft. lbs. for 3/8-16 bolts). Tighten braces first, then rails. When the assembly is complete, bolt the frames to floor.

6. Place insulating rail covers (rubber mats) onto the rails (if not already installed.)

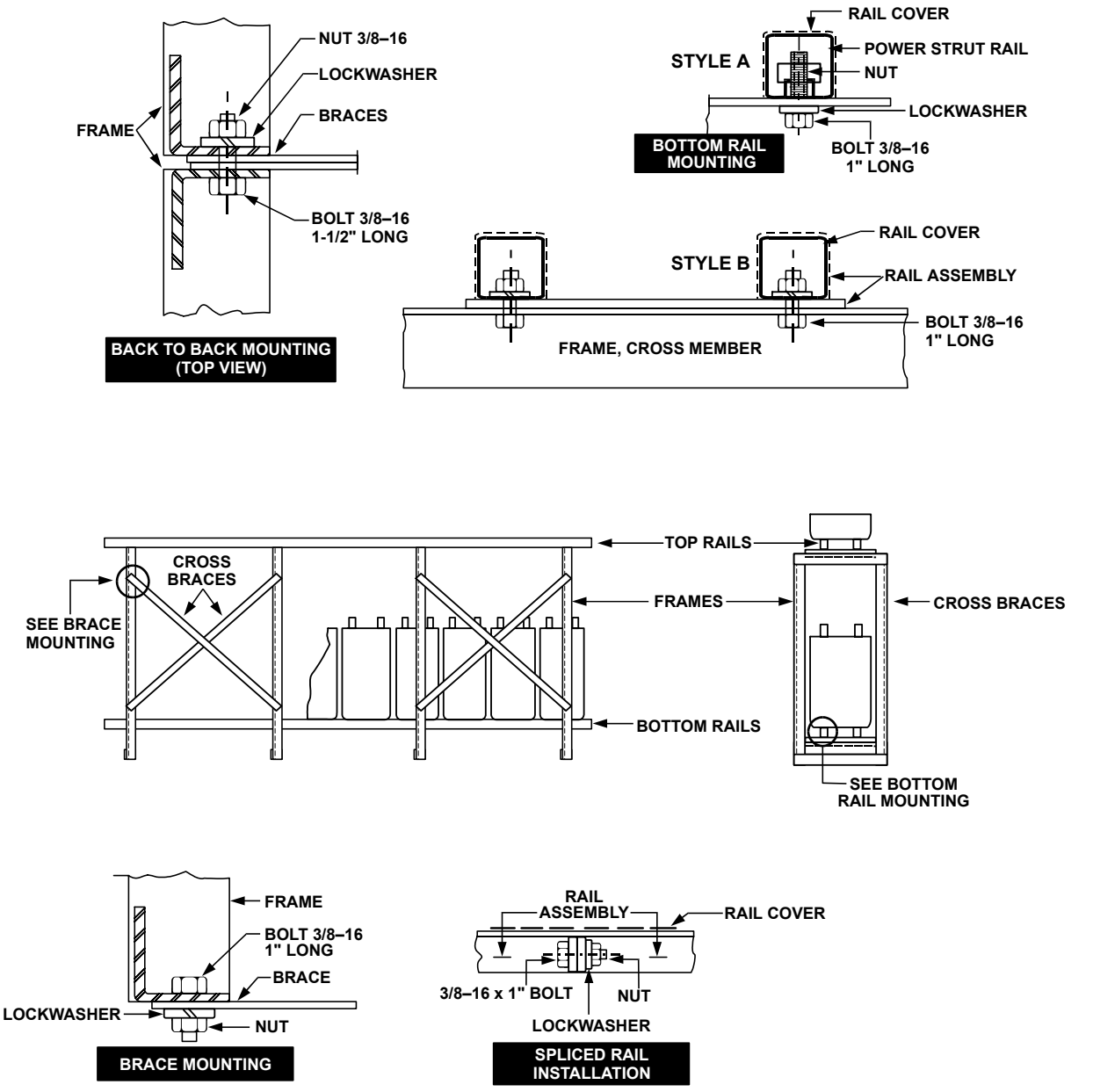



**Figure 6. Standard Racks – Non-Seismic**

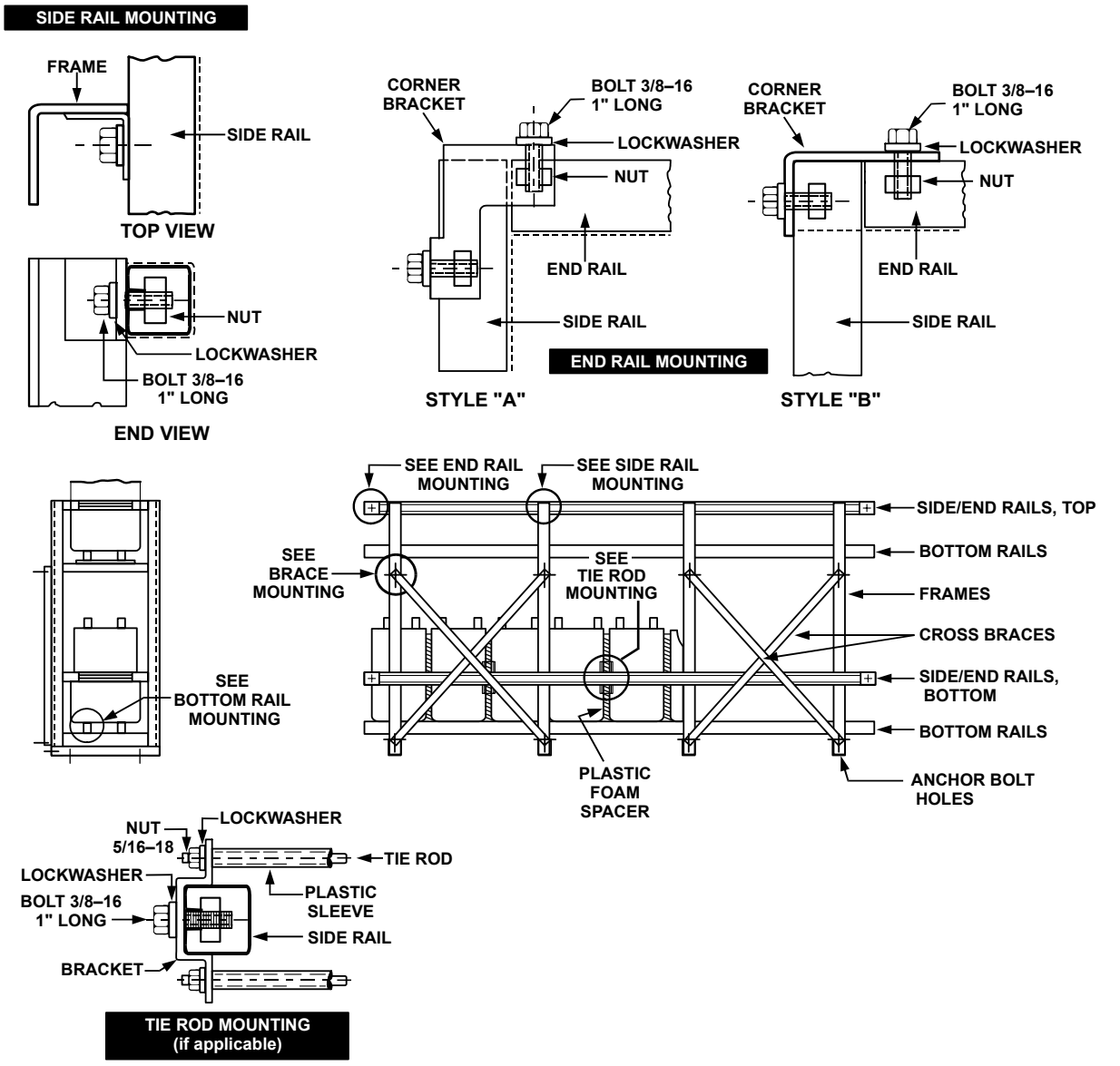



**Figure 7. Earthquake-Protected (EP) Racks**

### Battery Installation

Review the Battery Arrangement Outline drawing (MLI 1207) and note the weight of a single jar (4-cell unit). The outline drawing indicates which battery type is applicable via the Group number (G001 or G002).

Each of the 14 jars must be lifted over the 15 inch channel and placed on the battery rack. (Figs. 8, 9 and 10). A lifting device (such as a counter balance lift, floor crane or engine hoist) capable of reaching into the battery compartment is recommended. (Figs. 11 and 12).

**Note:** The number of battery jars will vary according to the size of the gas turbine unit, and the whether the battery system is sized for one or two units.

To facilitate exact positioning of cell jars, lubricate the top surface of the rubber mat with potable water. (See Figs. 13, 14, 15 and 16).

**Note:** **Do not use** grease, oil, solvents or any other substance which will damage or destroy the battery.

### Arrangement of Cells

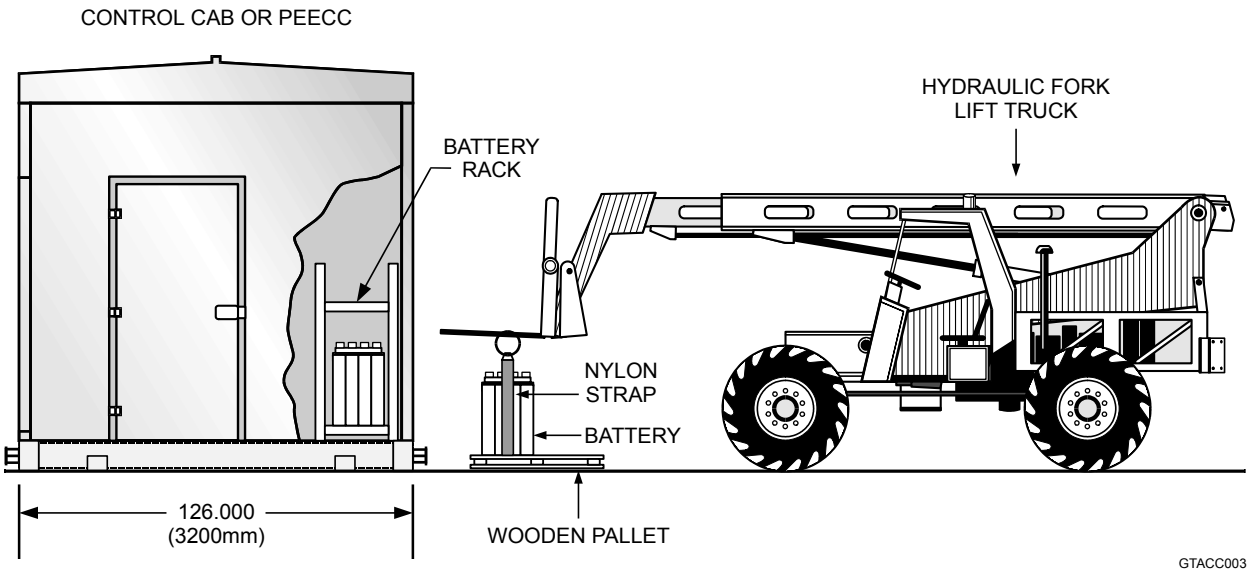
In tiered arrangements, install the first cells on the bottom tier of the rack to provide stability. In stepped arrangements, it may be more convenient to install the first cells on the top step of the rack rather than on the bottom step. This will avoid moving subsequent cells over those already placed.

Work from the center of the rack out towards both ends. Maintain a 1/2 inch space between cells. Arrange cells on the rack so that the positive post (terminal) of the next cell can be connected to the negative post (terminal) of the next cell. Cell post polarity is marked with symbols molded on the cover. A plus sign (+) denotes the positive post and a minus sign (-) the negative post.

When a rack drawing showing cell arrangement is supplied with the battery, be sure to follow it for optimum cabling efficiency. Improper cell arrangement will result in some cables being too long and some not being long enough.

Install tie rods (if required), to the battery rack. The tightening of tie rods must not apply side rail pressure to the cells. Nuts used with tie rods should only be tightened to about 3-5 ft. lbs. torque. To test this installation, an index card should be inserted between the side rails and cell container. (See Figs. 7 and 17).



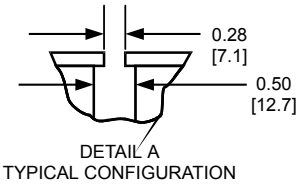




**NOTE:** Battery system location for "H" class and single shaft (STAG) gas turbines is on the ground floor of the Turbine Building.

**Figure 8. Lifting and Placement of Batteries**

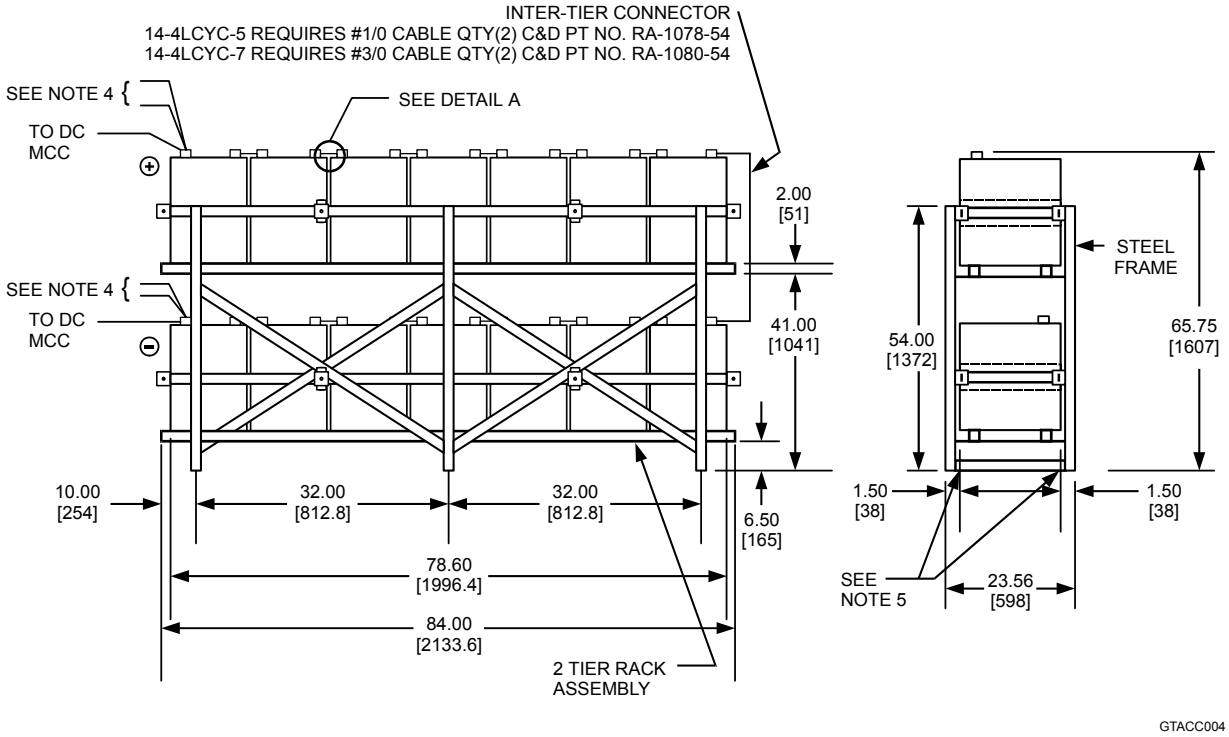
**NOTES:**

1. FOR MOUNTING OF C&D BATTERY TYPE 14-4LCY-5 114 BATTERY UNITS, 4 CELL TYPES PER UNIT, CELL TYPE LCY, 5 PLATES PER CELL, OR 14-4LCY-7 (7 PLATES PER CELL).
2. SEE BATTERY INSTALLATION INSTRUCTION DRAWING (MLI 0330) FOR ADDITIONAL INFORMATION.
3. SPACERS SHIPPED WITH BATTERY.
4. CHARGER(S) PREFERRED CONNECTION ARRANGEMENT AS SHOWN. ONE (1) POSITIVE (+) WIRE AND ONE (1) NEGATIVE (-) WIRE PER CHARGER.
5. FLOOR MOUNTING HOLES ARE .687 (17.4) ϕ (TWO IN EACH FRAME).
6. FLOOR MOUNTING BOLTS ARE NOT FURNISHED FOR CUSTOMER MOUNTED RACKS. (IF REQUIRED, INSTALLATION ANCHOR BOLT SIZE AND TYPE ARE TO BE DETERMINED BY USER IN ACCORDANCE WITH BUILDING CODES APPLICABLE TO THE ARRANGEMENT DESCRIBED BY THIS DRAWING).



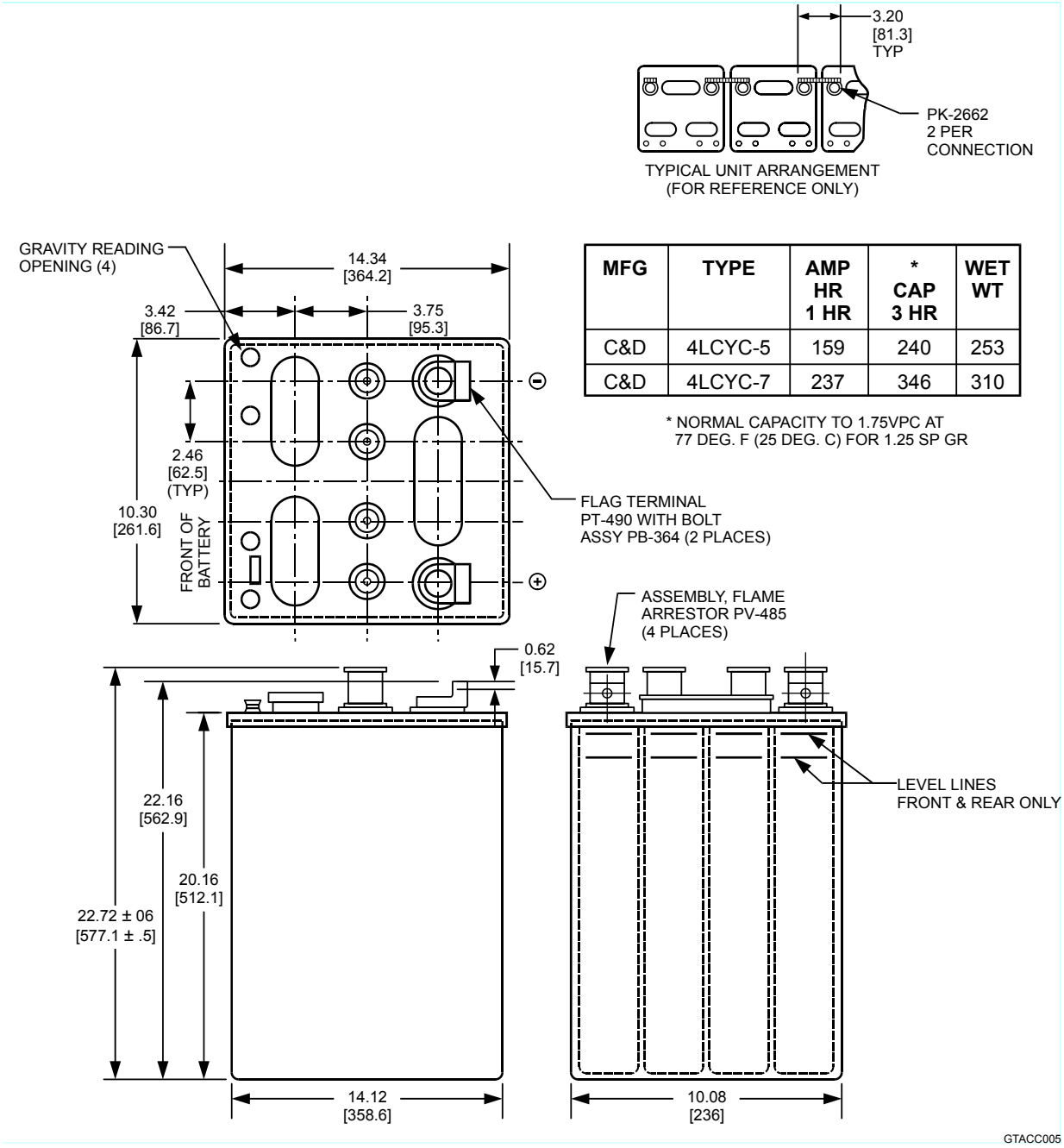

TYPICAL CONFIGURATION

| GROUP # | BATTERY CELL TYPE | RACK WEIGHT | BATTERY WEIGHT (PER UNIT) | TOTAL WEIGHT (14 UNITS & RACK) |
|---|---|---|---|---|
| G1 | 14-4LCYC-5 | 346 LBS [157 KGS] | 253 LBS [115 KGS] | 3888 LBS [1764 KGS] |
| G2 | 14-4LCYC-7 | 346 LBS [157 KGS] | 310 LBS [141 KGS] | 4686 LBS [2126 KGS] |




GTACC004

**Figure 9. Battery Rack Arrangement – Typical**




| MFG | TYPE | AMP HR 1 HR | * CAP 3 HR | WET WT |
|---|---|---|---|---|
| C&D | 4LCYC-5 | 159 | 240 | 253 |
| C&D | 4LCYC-7 | 237 | 346 | 310 |

* NORMAL CAPACITY TO 1.75VPC AT
77 DEG. F (25 DEG. C) FOR 1.25 SP GR

**Figure 10. Battery Cell Details – Typical**

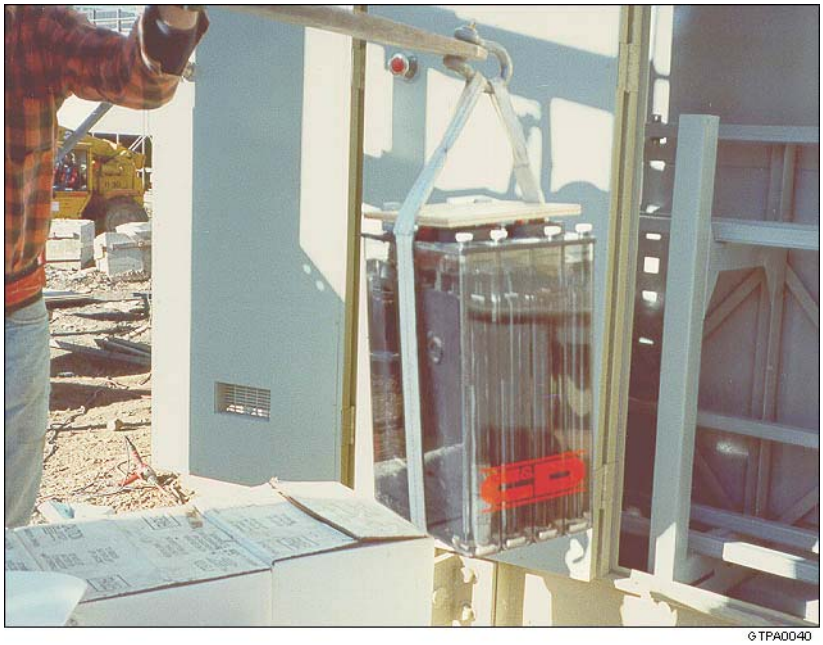



**Figure 11. Moving Battery Into PEECC Enclosure**

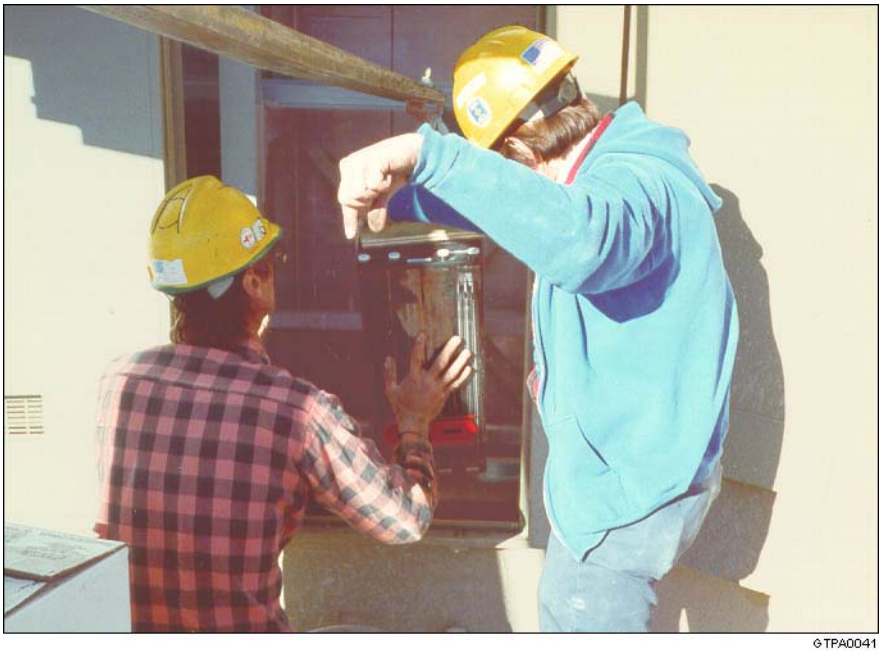



**Figure 12. Placing Battery in Rack**

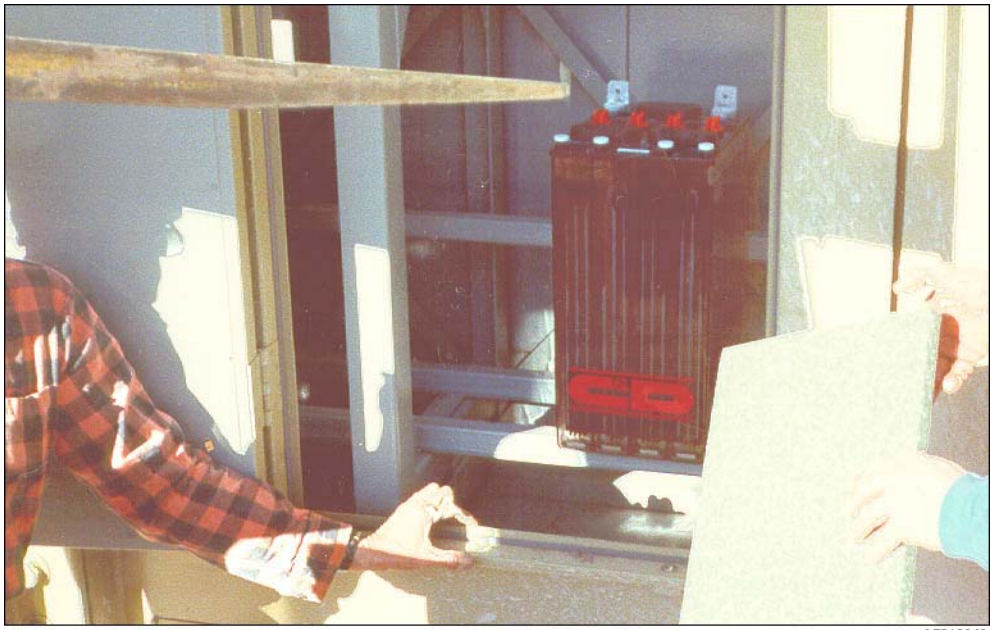



**Figure 13. Battery "Jar" Placed in Rack**

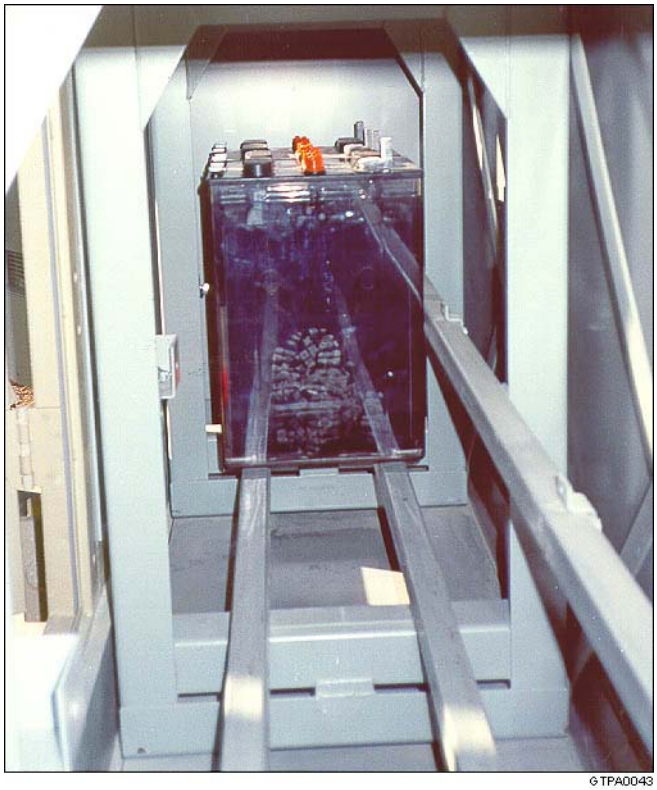



**Figure 14. Battery "Jars" In Position (End View)**



Figure 15. Batteries Set In Rack




Figure 16. Batteries Set In Rack

### Numbering of Cells

For ease of identification, all cells of a battery should be numbered. Plastic peel-and-stick numbers are furnished in the Accessories Package. The common practice is to start with "1" at the positive terminal of the battery and follow the electrical circuit with succeeding numbering. For a neat installation, first mark proposed number locations on cell jars with chalk. Then, remove plastic backing and firmly press each number into position. Be careful not to scratch the plastic jar.

**Note:** Do not use solvents to remove the original chalk mark.

### Flame Arresters

All C&D standby cells use flame arresters. On small cells they are factory installed. On large cells they are shipped separately and field installed. The large cells are shipped with orange colored vent plugs which must be removed and discarded before installing the flame arresters.

**WARNING: Dry-charged cells must be stored with orange vent caps and plastic film in place until ready for activation.**

**CAUTION: Be sure flame arresters are installed before making battery connections.**

### As Received Readings

Create a report form labeled “As Received Battery Readings”. Show date, turbine number, user customer name, site name, site location and Design Memo (DM) number in the heading. See Appendix A, Fig. A-1).

Select a cell and designate as the pilot cell. The pilot cell, selected for the purpose of the initial charge, is a temporary pilot cell. It should be the cell with the lowest specific gravity.

If a cell voltage is less than 2.10 volts, or the specific gravity (corrected to 77 degrees F) is less than 1.235, contact Gas Turbine Design Engineering.

### Battery Connections

Install the intercell connections and connect the battery to the battery charger(s). Refer to C&D Manual 12-800 Installing Cells section for the procedure.

Verify that the AC supply circuit breaker for each charger is open, and that the DC output circuit breaker for each charger is open before making any connections. (See Fig. 18).

**Note:** Do not connect the battery charger to the AC source (MCC) at this time.

**Initial Charge**

With the battery charger AC supply circuit breaker open, set the charger to “Float”, then close the battery charger DC circuit breaker. The charger DC voltmeter should indicate the battery voltage.

Close the battery charger AC circuit breaker. (The charger DC output current will build up gradually). If redundant chargers are supplied, repeat this step for the second charger.

Monitor the temperature of the electrolyte in the pilot cell. A thermometer should be held in the electrolyte for at least two minutes before taking the reading. To take temperature readings remove flame arrester cover, and insert the alcohol-type thermometer supplied with battery system.

Cell temperatures higher than 110 degrees F are not permissible except during the last three hours of initial charge. At no time shall the electrolyte temperature exceed 120 degrees F. If it is apparent that the next reading will exceed these limits, interrupt the charge and allow the electrolyte to cool to about 95 degrees F before resuming the charge.

When the pilot cell voltage is greater than 2.18 volts, set the charger to “Equalize”. If there are two chargers. Repeat for each charger. (See notes under Charger Settings).

Guard against electrolyte overflow. If it should become necessary to remove electrolyte to prevent overflowing, record removals and retain the electrolyte for possible reuse.

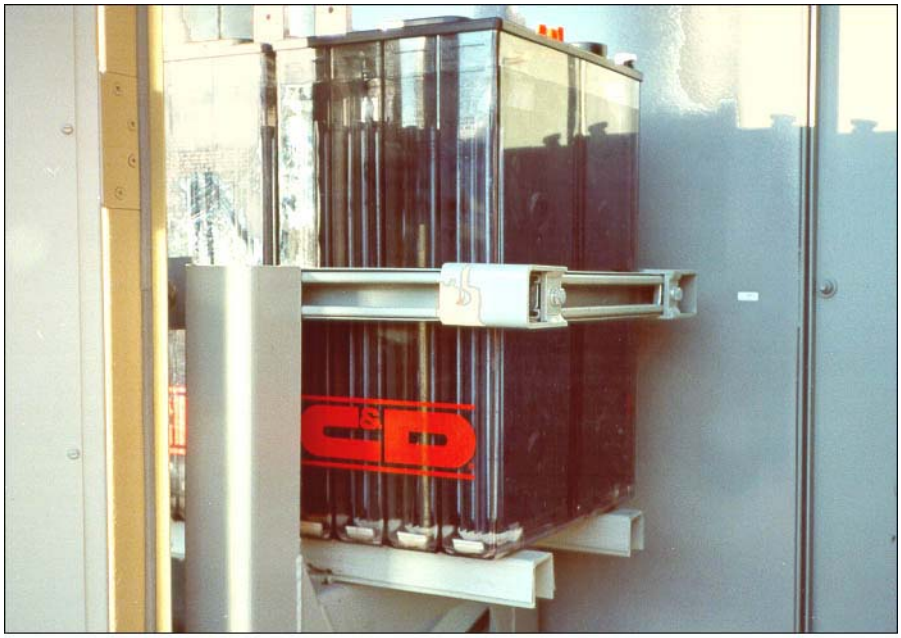



**Figure 17. Battery Rack Side Rail**




**Figure 18. Typical Cabling Arrangement**

### Charger Settings

Check the battery charger(s) float, equalize settings and adjust if necessary. If redundant chargers are supplied, disconnect the charger not being checked. See C&D Publication RS-759 (ARE Series Single-Phase Input), or other applicable publication for complete information.

With the battery on float charge and the charger not in the current limit mode, the battery voltage measured at the battery terminals should be 129 +/- 0.5 Volts DC.

With the battery on equalize charge and the charger not in the current limit mode, the battery voltage measured at the battery terminals should be 135 +/- 0.5 Volts DC.

**Note:** When redundant chargers with load sharing are provided, both chargers must be set to “Float” (or both set to “Equalize”) charge while both chargers are operating and connected to the battery.

To switch from float operation to equalize operation: Open the AC supply circuit breaker of the redundant charger and set both chargers to "Equalize” before re-closing the AC supply breaker of the redundant charger.

### Initial Charger Readings

When the pilot cell voltage reaches 2.4 +/- 0.02 volts, set the charger to “Float”. (When there are two chargers. (See notes under Charger Settings).

Create a report labeled “Initial Charger Battery Readings”. Show date, turbine number, user customer name, site name, site location and Design Memo (DM) number in the heading. (See Appendix A, Fig. A-2).

For each of the 56 cells (typical for the 7F series Gas Turbines), record voltage (+/- 0.005 V), electrolyte temperature (+/- 0.5 degrees F), electrolyte level (inches above low level mark (+/- 1/32 inch), and specific gravity (per the Measuring Specific Gravity section of the C&D Service Manual). Remove the flame arrester cover to measure temperature and to return electrolyte to the cells. Use an alcohol-type thermometer supplied with the batteries to measure cell temperature.

The report should include notes regarding any assumptions or conditions which might effect the readings. Place a copy of the report in the charger cabinet, and send another copy to the Customer and GE Field Office.

### Equalizing

The battery requires an “Equalizing” charge if it receives a deep discharge such that the pilot cell voltage is below 2.18 volts with the battery on float charge.

Disconnect all DC loads (especially the turbine control panel) before setting the charger to “Equalize“. If there are two chargers, see the notes under Charger Settings.

### As-Commissioned Readings

When the turbine installation and startup/commissioning is complete, a third set of battery readings should be taken. The report form should be similar to the initial charge readings form. See Appendix A, Fig. A-3).

### Battery Room Equipment ("H" Class Advanced Gas Turbines)

Pallets of batteries will be shipped to the job site as MSD Equipment. The 125V batteries will be installed into metal racks located in the Turbine Building on the Ground Floor under the control room. Installer will be responsible for assembling the metal racks, installing all of the batteries, installing the battery chargers, distribution panel, battery cables and wiring and all other battery room equipment. Refer to DN 523 (MS9001H Advanced Gas Turbine Installation Guide). See Figs. 19 and 20.




**Figure 19. Batteries Set In Metal Racks (Similar to MS9001H Plant Designs)**

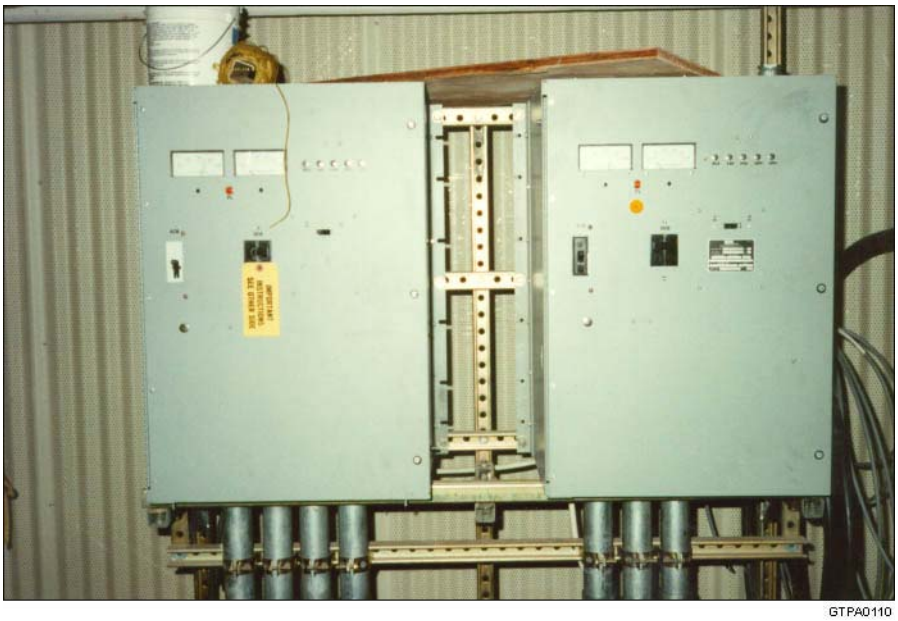



**Figure 20. Battery Chargers Mounted On Supports in Battery Room**

# Appendix A

## Contents

Figure. A-1 — As-Received Battery Readings Report

Figure. A-2 — Initial Charge Battery Readings Report

Figure. A-3 — As-Commissioned Battery Readings Report

## REPORT FORM NO. 1

***AS RECEIVED BATTERY READINGS***

**BATTERY TYPE** _____ **DATE** _____
**TURBINE S/N** _____ **DM NO.** _____
**CUSTOMER/SITE** _____
**COMMENTS** _____

| Cell # | Volts +2.000 | F | SP Gravity +1.000 | | Cell # | Volts +2.000 | F | SP Gravity +1.000 |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | 29 | | | |
| 2 | | | | | 30 | | | |
| 3 | | | | | 31 | | | |
| 4 | | | | | 32 | | | |
| 5 | | | | | 33 | | | |
| 6 | | | | | 34 | | | |
| 7 | | | | | 35 | | | |
| 8 | | | | | 36 | | | |
| 9 | | | | | 37 | | | |
| 10 | | | | | 38 | | | |
| 11 | | | | | 39 | | | |
| 12 | | | | | 40 | | | |
| 13 | | | | | 41 | | | |
| 14 | | | | | 42 | | | |
| 15 | | | | | 43 | | | |
| 16 | | | | | 44 | | | |
| 17 | | | | | 45 | | | |
| 18 | | | | | 46 | | | |
| 19 | | | | | 47 | | | |
| 20 | | | | | 48 | | | |
| 21 | | | | | 49 | | | |
| 22 | | | | | 50 | | | |
| 23 | | | | | 51 | | | |
| 24 | | | | | 52 | | | |
| 25 | | | | | 53 | | | |
| 26 | | | | | 54 | | | |
| 27 | | | | | 55 | | | |
| 28 | | | | | 56 | | | |

**Figure. A-1 — As-Received Battery Readings Report Form**

## REPORT FORM NO. 2

***INITIAL CHARGE BATTERY READINGS***

**BATTERY TYPE** _____ **DATE** _____
**TURBINE S/N** _____ **DM NO.** _____
**CUSTOMER/SITE** _____
**COMMENTS** _____

| Cell # | Volts +2.000 | F | SP Gravity +1.000 | Elyt Lvl. | | Cell # | Volts +2.000 | F | SP Gravity +1.000 | Elyt Lvl. |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | 29 | | | | |
| 2 | | | | | | 30 | | | | |
| 3 | | | | | | 31 | | | | |
| 4 | | | | | | 32 | | | | |
| 5 | | | | | | 33 | | | | |
| 6 | | | | | | 34 | | | | |
| 7 | | | | | | 35 | | | | |
| 8 | | | | | | 36 | | | | |
| 9 | | | | | | 37 | | | | |
| 10 | | | | | | 38 | | | | |
| 11 | | | | | | 39 | | | | |
| 12 | | | | | | 40 | | | | |
| 13 | | | | | | 41 | | | | |
| 14 | | | | | | 42 | | | | |
| 15 | | | | | | 43 | | | | |
| 16 | | | | | | 44 | | | | |
| 17 | | | | | | 45 | | | | |
| 18 | | | | | | 46 | | | | |
| 19 | | | | | | 47 | | | | |
| 20 | | | | | | 48 | | | | |
| 21 | | | | | | 49 | | | | |
| 22 | | | | | | 50 | | | | |
| 23 | | | | | | 51 | | | | |
| 24 | | | | | | 52 | | | | |
| 25 | | | | | | 53 | | | | |
| 26 | | | | | | 54 | | | | |
| 27 | | | | | | 55 | | | | |
| 28 | | | | | | 56 | | | | |

**Figure. A-2 — Initial Charge Battery Readings Report Form**

## REPORT FORM NO. 3

***AS COMMISSIONED BATTERY READINGS***

**BATTERY TYPE** _____ **DATE** _____
**TURBINE S/N** _____ **DM NO.** _____
**CUSTOMER/SITE** _____
**COMMENTS** _____

| Cell # | Volts +2.000 | F | SP Gravity +1.000 | Elyt Lvl. | | Cell # | Volts +2.000 | F | SP Gravity +1.000 | Elyt Lvl. |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | 29 | | | | |
| 2 | | | | | | 30 | | | | |
| 3 | | | | | | 31 | | | | |
| 4 | | | | | | 32 | | | | |
| 5 | | | | | | 33 | | | | |
| 6 | | | | | | 34 | | | | |
| 7 | | | | | | 35 | | | | |
| 8 | | | | | | 36 | | | | |
| 9 | | | | | | 37 | | | | |
| 10 | | | | | | 38 | | | | |
| 11 | | | | | | 39 | | | | |
| 12 | | | | | | 40 | | | | |
| 13 | | | | | | 41 | | | | |
| 14 | | | | | | 42 | | | | |
| 15 | | | | | | 43 | | | | |
| 16 | | | | | | 44 | | | | |
| 17 | | | | | | 45 | | | | |
| 18 | | | | | | 46 | | | | |
| 19 | | | | | | 47 | | | | |
| 20 | | | | | | 48 | | | | |
| 21 | | | | | | 49 | | | | |
| 22 | | | | | | 50 | | | | |
| 23 | | | | | | 51 | | | | |
| 24 | | | | | | 52 | | | | |
| 25 | | | | | | 53 | | | | |
| 26 | | | | | | 54 | | | | |
| 27 | | | | | | 55 | | | | |
| 28 | | | | | | 56 | | | | |

**Figure. A-3 — As-Commissioned Battery Readings Report Form**